

## Use Case Descriptions

|                                                 |                                                                                                                                                                                                                                                             |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case 1</b>                               | <b>Register an account</b>                                                                                                                                                                                                                                  |
| <b>Summary</b>                                  | Create an account; system sends a verification email.                                                                                                                                                                                                       |
| <b>Actor</b>                                    | User                                                                                                                                                                                                                                                        |
| <b>Precondition</b>                             | User owns an email address.                                                                                                                                                                                                                                 |
| <b>Postcondition</b>                            | Inactive account exists; verification email sent.                                                                                                                                                                                                           |
| <b>Base sequence</b>                            | <ol style="list-style-type: none"><li>1. User enters email &amp; password [ALT1][ALT2].</li><li>2. System creates an inactive account.</li><li>3. System sends a time-limited verification email [ALT3].</li><li>4. System confirms registration.</li></ol> |
| <b>Alternate sequence (branch or exception)</b> | [ALT1] Email already registered.<br>[ALT2] Invalid email or weak password.<br>[ALT3] Email delivery issue.<br>[*ALT0] Can cancel at any time.                                                                                                               |
| <b>Note</b>                                     | Covers requirement 1                                                                                                                                                                                                                                        |

|                                                 |                                                                                                                                                                                       |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case 2</b>                               | <b>Verify email</b>                                                                                                                                                                   |
| <b>Summary</b>                                  | Activate an account via verification link.                                                                                                                                            |
| <b>Actor</b>                                    | User                                                                                                                                                                                  |
| <b>Precondition</b>                             | Inactive account; valid verification token issued.                                                                                                                                    |
| <b>Postcondition</b>                            | Account marked verified.                                                                                                                                                              |
| <b>Base sequence</b>                            | <ol style="list-style-type: none"> <li>1. User follows email link.</li> <li>2. System validates token [ALT1].</li> <li>3. System marks account verified and shows success.</li> </ol> |
| <b>Alternate sequence (branch or exception)</b> | [ALT1] Token expired/invalid/already used.<br>[*ALT0] Can cancel at any time.                                                                                                         |
| <b>Note</b>                                     | Covers requirement 2                                                                                                                                                                  |

|                                                 |                                                                                                                                                                                |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case 3</b>                               | <b>Log in</b>                                                                                                                                                                  |
| <b>Summary</b>                                  | Authenticate to access the application.                                                                                                                                        |
| <b>Actor</b>                                    | User                                                                                                                                                                           |
| <b>Precondition</b>                             | Account exists and is verified.                                                                                                                                                |
| <b>Postcondition</b>                            | Authentication session established.                                                                                                                                            |
| <b>Base sequence</b>                            | <ol style="list-style-type: none"> <li>1. User logs in entering email &amp; password [ALT1].</li> <li>2. System validates credentials.</li> <li>3. Session created.</li> </ol> |
| <b>Alternate sequence (branch or exception)</b> | [ALT1] Wrong credentials (go back to <b>1</b> ).<br>[*ALT0] Can cancel at any time.                                                                                            |
| <b>Note</b>                                     | Covers requirement 3                                                                                                                                                           |

|                                                 |                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case 4</b>                               | <b>Reset password</b>                                                                                                                                                                                                                                                                                                                               |
| <b>Summary</b>                                  | User resets forgotten password.                                                                                                                                                                                                                                                                                                                     |
| <b>Actor</b>                                    | User                                                                                                                                                                                                                                                                                                                                                |
| <b>Precondition</b>                             | User has an active account.                                                                                                                                                                                                                                                                                                                         |
| <b>Postcondition</b>                            | User regains access of the account.                                                                                                                                                                                                                                                                                                                 |
| <b>Base sequence</b>                            | <ol style="list-style-type: none"> <li>1. User inserts email address.</li> <li>2. System validates email address [ALT1].</li> <li>3. System sends recovery email to user's email address.</li> <li>4. User inserts new password.</li> <li>5. System validates password [ALT2].</li> <li>6. System assigns new password to email address.</li> </ol> |
| <b>Alternate sequence (branch or exception)</b> | <p>[ALT1] Email doesn't exist/invalid.</p> <ul style="list-style-type: none"> <li>• Go back to <b>1</b>.</li> </ul> <p>[ALT2] Weak password.</p> <ul style="list-style-type: none"> <li>• Go back to <b>4</b>.</li> </ul> <p>[*ALT0] Can cancel at any time.</p>                                                                                    |
| <b>Note</b>                                     | Covers requirement 4                                                                                                                                                                                                                                                                                                                                |

|                                                 |                                                                                                                                |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case 5</b>                               | <b>Log out</b>                                                                                                                 |
| <b>Summary</b>                                  | End the authenticated session.                                                                                                 |
| <b>Actor</b>                                    | User                                                                                                                           |
| <b>Precondition</b>                             | Authenticated session exists.                                                                                                  |
| <b>Postcondition</b>                            | Session revoked.                                                                                                               |
| <b>Base sequence</b>                            | <ol style="list-style-type: none"> <li>1. User logs out.</li> <li>2. System invalidates session and clears cookies.</li> </ol> |
| <b>Alternate sequence (branch or exception)</b> | [*ALT0] Can cancel at any time.                                                                                                |
| <b>Note</b>                                     | Covers requirement 5                                                                                                           |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case 6</b>                               | <b>Link provider's account</b>                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Summary</b>                                  | Link Google Drive/ OneDrive for read-only access.                                                                                                                                                                                                                                                                                                                                                           |
| <b>Actor</b>                                    | User                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Precondition</b>                             | User is logged in.                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Postcondition</b>                            | Google/Microsoft tokens stored securely; status Connected.                                                                                                                                                                                                                                                                                                                                                  |
| <b>Base sequence</b>                            | <ol style="list-style-type: none"> <li>1. User selects provider.</li> <li>2. User connects to chosen provider.</li> <li>3. System redirects to provider's consent for read only access [ALT1].</li> <li>4. User consents; provider redirects back with code [ALT2].</li> <li>5. System exchanges code for tokens; stores encrypted [ALT3].</li> <li>6. System confirms connection with provider.</li> </ol> |
| <b>Alternate sequence (branch or exception)</b> | <p>[ALT1] User denies consent (go back to 1).</p> <p>[ALT2] Code missing/invalid (go back to 1).</p> <p>[ALT3] Token exchange fails (go back to 1).</p> <p>[*ALT0] Can cancel at any time.</p>                                                                                                                                                                                                              |
| <b>Note</b>                                     | Covers requirement 6                                                                                                                                                                                                                                                                                                                                                                                        |

|                                          |                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                  |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Use case 7                               | Manage provider connection (status / refresh / disconnect)                                                                                                                                                                                                                                                                           |                                                                                                                                  |
| Summary                                  | View connection state, auto-refresh tokens when needed, or disconnect.                                                                                                                                                                                                                                                               |                                                                                                                                  |
| Actor                                    | User                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                  |
| Precondition                             | At least one provider previously linked.                                                                                                                                                                                                                                                                                             |                                                                                                                                  |
| Postcondition                            | Status displayed; token may be refreshed; provider may be disconnected.                                                                                                                                                                                                                                                              |                                                                                                                                  |
| Base sequence                            | Scenario A<br>View connection state                                                                                                                                                                                                                                                                                                  | 1. User opens Connections.<br>2. System lists the providers with connection state.                                               |
|                                          | Scenario B<br>Auto-refresh tokens when needed                                                                                                                                                                                                                                                                                        | 1. User performs an action that calls a provider API.<br>2. If access token expired, the system attempts to refresh once [ALT1]. |
|                                          | Scenario C<br>Disconnect from provider                                                                                                                                                                                                                                                                                               | 1. User clicks disconnect for a provider.<br>2. System deletes tokens.<br>3. Status becomes not connected [ALT2].                |
| Alternate sequence (branch or exception) | <p>[ALT1] Refresh failed (revoked/expired).</p> <ul style="list-style-type: none"><li>• Show reconnect.</li></ul> <p>[ALT2] Delete fails (no tokens found).</p> <ul style="list-style-type: none"><li>• Show success but allow reconnect later.</li></ul> <p>[ALT3] Email delivery issue.</p> <p>[*ALT0] Can cancel at any time.</p> |                                                                                                                                  |
| Note                                     | Covers requirement 7, 8, 9                                                                                                                                                                                                                                                                                                           |                                                                                                                                  |

|                                                 |                                                                                                                                                        |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case 8</b>                               | <b>Browse unified files (read-only)</b>                                                                                                                |
| <b>Summary</b>                                  | Browse a single dashboard for all connected providers.                                                                                                 |
| <b>Actor</b>                                    | User                                                                                                                                                   |
| <b>Precondition</b>                             | At least one provider connected.                                                                                                                       |
| <b>Postcondition</b>                            | Read-only listing shown                                                                                                                                |
| <b>Base sequence</b>                            | <ol style="list-style-type: none"> <li>1. User wants to see all files.</li> <li>2. System loads first page merged from connected providers.</li> </ol> |
| <b>Alternate sequence (branch or exception)</b> | [*ALT0] Can cancel at any time.                                                                                                                        |
| <b>Note</b>                                     | Covers requirement 10                                                                                                                                  |



|                                                 |                                                                                                                                                                                                                                                   |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case 9</b>                               | <b>Cross-provider search</b>                                                                                                                                                                                                                      |
| <b>Summary</b>                                  | Search by name across all connected providers.                                                                                                                                                                                                    |
| <b>Actor</b>                                    | User                                                                                                                                                                                                                                              |
| <b>Precondition</b>                             | At least one provider connected                                                                                                                                                                                                                   |
| <b>Postcondition</b>                            | Results displayed                                                                                                                                                                                                                                 |
| <b>Base sequence</b>                            | <ol style="list-style-type: none"> <li>1. User enters file name</li> <li>2. User selects one or both providers</li> <li>3. System queries only selected providers</li> <li>4. System aggregates and displays results<br/>[ALT1] [ALT2]</li> </ol> |
| <b>Alternate sequence (branch or exception)</b> | <p>[ALT1] Provider error.</p> <ul style="list-style-type: none"> <li>• Retry or reconnect provider.</li> </ul> <p>[ALT2] No matches.</p> <ul style="list-style-type: none"> <li>• Go back to 1.</li> </ul> <p>[*ALT0] Can cancel at any time.</p> |
| <b>Note</b>                                     | Covers requirement 11                                                                                                                                                                                                                             |

|                                                 |                                                                                                                                        |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case 10</b>                              | <b>See item's metadata</b>                                                                                                             |
| <b>Summary</b>                                  | View the selected item's metadata                                                                                                      |
| <b>Actor</b>                                    | User                                                                                                                                   |
| <b>Precondition</b>                             | Item exists; user has permission; provider reachable.                                                                                  |
| <b>Postcondition</b>                            | Metadata of the selected item visible.                                                                                                 |
| <b>Base sequence</b>                            | <ol style="list-style-type: none"> <li>1. User selects item to view its meta data.</li> <li>2. System shows items metadata.</li> </ol> |
| <b>Alternate sequence (branch or exception)</b> | [*ALT0] Can cancel at any time.                                                                                                        |
| <b>Note</b>                                     | Covers requirement 12                                                                                                                  |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case 11</b>                              | <b>Open item in provider</b>                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Summary</b>                                  | Open the selected item in its native provider web app.                                                                                                                                                                                                                                                                                                                                              |
| <b>Actor</b>                                    | User                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Precondition</b>                             | Item exists; user has permission; provider reachable.                                                                                                                                                                                                                                                                                                                                               |
| <b>Postcondition</b>                            | Provider viewer opened in a new tab.                                                                                                                                                                                                                                                                                                                                                                |
| <b>Base sequence</b>                            | <ol style="list-style-type: none"> <li>1. User selects item to open.</li> <li>2. System opens the item's URL in the providers web app (in a new tab) [ALT1][ALT2].</li> </ol>                                                                                                                                                                                                                       |
| <b>Alternate sequence (branch or exception)</b> | <p>[ALT1] Token expired.</p> <ul style="list-style-type: none"> <li>• Attempt refresh <ul style="list-style-type: none"> <li>○ On failure show reconnect</li> </ul> </li> </ul> <p>[ALT2] Permission denied/not found.</p> <ul style="list-style-type: none"> <li>• Show error <ul style="list-style-type: none"> <li>○ Go back to 1.</li> </ul> </li> </ul> <p>[*ALT0] Can cancel at any time.</p> |
| <b>Note</b>                                     | Covers requirement 13                                                                                                                                                                                                                                                                                                                                                                               |